Adam Ben

adam.ben55@proton.me | +44 78 65 14 67 43 | +33 7 49 71 17 40 | Metz, France | Full Driving Licence
A7336.github.io | adamb7336 | Full Right to Work in the UK & EU

Computer Science student proficient in Python, Java, and C with hands-on experience building asynchronous pipelines, game AI engines, and containerized systems.

EDUCATION

BSc in Computer Science 09/2024 – Present
Université de Lorraine Metz, France
- **Academic Ranking:** Class Valedictorian (Ranked 1st out of 72 students)
- **Final Grade:** 18.5/20 (First-Class Honours equivalent)

PROJECTS

Real-Time London Underground Monitoring Platform 06/2026 – 08/2026
Python, FastAPI, WebSockets, PostgreSQL, Docker, Streamlit
- Built an asynchronous FastAPI pipeline (httpx/asyncio) to ingest TfL REST API feeds into PostgreSQL 15, optimizing historical query performance with DISTINCT ON indexing.
- Engineered a differential WebSocket system to push updates only during state changes, and containerized the multi-service architecture using multi-stage Docker builds.

Breakthrough Game AI Engine (Symbolic AI) 01/2026 – 06/2026
Java, Game Theory, Metaheuristics, MCTS
- Developed a simplified chess engine in Java featuring Minimax with Alpha-Beta Pruning and Monte Carlo Tree Search (MCTS) for statistical move evaluation.
- Automated heuristic function calibration by implementing tournament-based Genetic Algorithms and Simulated Annealing metaheuristics.

Multiplayer Distributed Clue Game 01/2026 – 06/2026
Java, JavaFX, Sockets, SQL, JUnit
- Built a distributed multiplayer game in Java/JavaFX with real-time TCP socket networking, encrypted data transmissions, and an SQL database backend.
- Implemented dynamic feature-toggling via custom XML configurations and verified client-server component reliability using JUnit test suites.

SKILLS

Languages
C, C++, Java, Python, SQL, PHP

Backend
FastAPI, WebSockets, SQLAlchemy, REST APIs, JavaFX, Streamlit

Databases & Tools
PostgreSQL, Docker, Git, JUnit

PROFESSIONAL EXPERIENCE

Warehouse Associate 06/2025 – 09/2025
Integrated Service Solutions Ltd. (ISS Ltd.) Sittingbourne
- Audited shipment labels for cross-border logistics to detect compliance errors early, preventing delivery delays and supply chain bottlenecks.
- Maintained 100% processing accuracy while handling 960+ units per shift under strict daily operational deadlines for enterprise clients (e.g., M&S, Tesco).

Server 06/2022 – 09/2022
Creams Cafe Gravesend
- Executed daily transactions using POS software, troubleshooting payment discrepancies and ensuring operational data accuracy.
- Coordinated real-time order queues between kitchen staff and delivery drivers during high-volume periods, earning management recognition for swift problem-solving.

LANGUAGES

English
Native

French
Native

Arabic
Native

INTERESTS

Building Computers | International Travel | Biking | Basketball